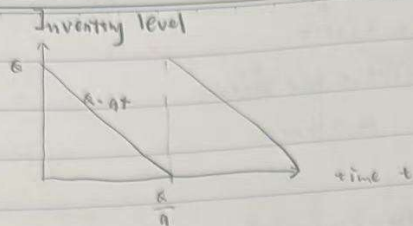


Question 1

(a)



Setup cost $k = 400$ purchasing unit per cycle Q/a
 Holding cost per cycle $h \cdot \frac{Q}{2} \cdot \frac{Q}{a} = \frac{Q^2 h}{2a}$, $a = 35000$ Litre/month

$$\text{Total Cost Per Cycle} = k + cQ + \frac{Q^2 h}{2a}$$

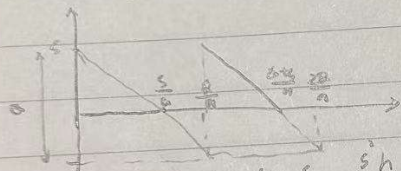
$$T(Q) = \frac{k + cQ + \frac{Q^2 h}{2a}}{Q/a} = \frac{ak}{Q} + ac + \frac{Qh}{2}$$

$$\text{Optimal } Q^* = \sqrt{\frac{2ak}{h}} = \sqrt{\frac{2 \times 35000 \times 400}{0.06}} \approx 21602 \text{ Litre}$$

$$\text{Order time } \frac{Q^*}{a} = \frac{21602}{35000} \approx 0.617 \text{ month}$$

Inventory level

(b)



$$\text{Holding cost} = h \cdot \frac{Q-s}{2} \cdot \frac{Q-s}{a} = \frac{(Q-s)^2 h}{2a}$$

$$\text{Total cost} = k + cQ + \frac{(Q-s)^2 h}{2a} + \frac{P(Q-s)^2}{2a}$$

$$T(Q, s) = \frac{ak}{Q} + ac + \frac{(Q-s)^2 h}{2a} + \frac{P(Q-s)^2}{2a}$$

$$Q^* = \sqrt{\frac{2ak}{h}} \cdot \sqrt{\frac{P+h}{P}} \approx 21602 \cdot \sqrt{\frac{0.06+0.25}{0.25}} \approx 23382 \text{ Litre}$$

$$t^* = \frac{Q^*}{a} \approx \frac{23382}{35000} \approx 0.668 \text{ month}$$

Question 1 (c)

$$c = \begin{cases} c_1 = 0.95 & Q < 1500 \\ c_2 = 0.81 & 1500 \leq Q < 4000 \\ c_3 = 0.78 & Q \geq 4000 \end{cases}$$

$$K = 400, \quad h = 0.06, \quad n = 3500$$

$$\bar{T}_j(Q) = \frac{nK}{Q} + nC_j + \frac{hQ}{2}, \quad \text{for } j = 1, 2, 3$$

$$EOQ = \sqrt{\frac{2nK}{h}} \approx 21602 \text{ litres}$$

$$EOQ \in [1500, 4000], \text{ corresponding } c_2 = 0.81$$

$$\bar{T}_2^* = \bar{T}_2(EOQ) = \frac{nK}{EOQ} + nC_2 + \frac{h \cdot EOQ}{2} \approx 29646$$

$$\bar{T}_3^* = \bar{T}_3(Q_2) = \frac{nK}{Q_2} + nC_3 + \frac{hQ_2}{2} = 28850$$

$$T^* = \min\{\bar{T}_2^*, \bar{T}_3^*\} = 28850, \quad Q^* = 4000 = Q_2, \quad t^* = \frac{40000}{3500} \approx 1.143 \text{ months}$$

Question 2 unit cost $c = 2000$, $k = 3000$, $h = 40$

$$r_1 = 30, r_2 = 40, r_3 = 25, r_4 = 35, r_5 = 30, r_6 = 40$$

$W = 90$ (The max holding amount)

assume: first is not, satisfies from week 2 to 6

$$\text{Total demand} = 30 + 40 + 25 + 35 + 30 + 40 = 190$$

$$\text{purchase cost} = 190 \times 2000 = 380000$$

Try to minimize purchase times: $\frac{3000}{40} = 75$, if storing amount < 75

storing is better than purchase.

$$r_1 + r_2 = 70 < 75, \quad r_1 + r_2 + r_3 = 95 > 75, \quad \text{so } r_1 + r_2 \text{ first time}$$

$$r_3 + r_4 = 60 < 75, \quad r_3 + r_4 + r_5 = 90 > 75, \quad \text{so } r_3 + r_4 \text{ second time}$$

$$r_5 + r_6 = 70 < 75, \quad \text{so } r_5 + r_6 \text{ third time}$$

optimal = week 1 = order $r_1 + r_2$

week 3 = order $r_3 + r_4$

week 5 = order $r_5 + r_6$

$$\begin{aligned} \text{Total cost (order)} &= [3000 + (40 \times 70)] + [3000 + (40 \times 60)] + [3000 + (40 \times 70)] \\ &= 13200 \end{aligned}$$

$$\therefore \text{Total cost (order + purchase)} = 13200 + 380000 = 393200$$

Question 3

(a) $c = 20$, $p = 32$, $h = 20 - 12 = 8$

Total cost $C(D, y)$

$= cy + h \max(0, y - D) + p \max(0, D - y)$

$F(y^*) = \frac{p - c}{p + h} = \frac{12}{40} = 0.3$

cdf. d	20	25	30
Φ	0.03	0.15	0.18
Φ	0.03	0.18	0.33

$\therefore P(0 \leq y^*) = \frac{p - c}{p + h}$

$\Rightarrow y^* = 30$

(b) $E[\max(0, D - y)] = \sum_{d \geq y} (d - y) P(D = d)$

$= (30 - 20) \times 0.03 + (40 - 20) \times 0.12 + (45 - 20) \times 0.18 + (50 - 20) \times 0.1$

$= 7.25$

$E[\max(0, y - D)] = \sum_{d < y} (y - d) P(D = d)$

$= (30 - 20) \times 0.03 + (30 - 25) \times 0.12 = 0.1$

(c) $E(\text{cost}) = cy + h E[\max(0, y - D)] + p E[\max(0, D - y)]$

$= 20 \times 30 + 8 \times 7.25 + 32 \times 0.1 = 839.2$

Question 4

(a) In 20 minutes:

$$\lambda_{20} = 20 \times 0.25 = 5 \text{ female/minutes}$$

$$P(X=6) = \frac{e^{-5} \cdot 5^6}{6!} = 0.1462$$

$$(b) P(X \leq 5) = C_0^5 \cdot (0.5)^6 + C_1^5 (0.5)^6$$

$$= 0.1094$$

(c) (A)

$$(c) \text{ Total Arrival Rate} = 0.15 + 0.25 = 0.4 \text{ person/min}$$

$$\text{Expected of ninth customer} = \frac{9}{0.4} = 22.5 \text{ mins}$$

$$(d) \mu = \frac{1}{2}, \lambda = 0.4/\text{min} \text{ inter/queue}$$

$$\rho = \frac{\lambda}{\mu} = 0.8$$

$$P_0 = 1 - \rho = 0.2$$

(e)

$$(e) L = \frac{\rho}{1 - \rho} = \frac{0.8}{0.2} = 4, \text{ average waiters number}$$

$$(f) W_q = \frac{\rho}{\mu(1 - \rho)} = \frac{0.8}{0.5 \times 0.2} = 8 \text{ mins}$$

$$(g) T_{\text{next, male}} = \text{Exp}(0.15) \quad T_{\text{next, female}} = \text{Exp}(0.25)$$

next is male: $T_{\text{next, male}} < T_{\text{next, female}}$

$$\therefore P(T_1 < T_2) = \frac{\lambda_1}{\lambda_1 + \lambda_2} = 0.375$$

Question 5

$$(a) \quad \lambda = \frac{1}{15} \text{ arrival / min}$$

$$\mu = \frac{8}{60} = \frac{2}{15} \text{ service / min}, \quad \rho = \frac{\lambda}{\mu} = \frac{1}{2}$$

$$\text{no replied : idle. } \pi_0 = 1 - \rho = \frac{1}{2}$$

$$(b) \quad L = \frac{\rho}{1 - \rho} = 1$$

$$(c) \quad W_q = \frac{\rho}{\mu(1 - \rho)} = \frac{\frac{1}{2}}{\frac{2}{15} \times \frac{1}{2}} = 7.5 \text{ min}$$